

Technical Documentation

“Charred Memory”

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System Overview

This project is a **2D Adventure Side-scroller** featuring pixel art aesthetics and elements of psychological horror. The system simulates a narrative-driven experience where the user controls a protagonist (a knight) awakening in a post-apocalyptic, burnt forest. The core functionality includes player movement, environmental interaction, and narrative progression involving NPCs (specifically a mysterious crow with red eyes). The game logic handles state transitions between "dream" and "reality" sequences.

Architecture

The application is built upon **Unity's Component-Based Architecture**:

- **Scene Management:** Utilizes a persistent architecture (using `DontDestroyOnLoad`) to maintain game state and UI settings across different scenes (Main Menu, Forest Level, etc.).
- **UI Controller (Singleton):** Implements a Singleton pattern for the `SettingsController` to manage global audio volume and pause states, ensuring the UI persists without duplication during gameplay transitions.
- **Physics System:** relies on Unity's built-in 2D Physics engine (`Rigidbody2D`, `BoxCollider2D`) for player movement and collision detection.

Technology Stack

- **Game Engine:** Unity 2022.3 (LTS) or higher.
- **Programming Language:** C# .
- **Rendering Pipeline:** Universal Render Pipeline (URP) 2D for handling advanced lighting effects, rain particles, and atmospheric gloom.
- **Version Control:** Git (hosted on GitHub) utilizing a Unity-specific `.gitignore` file to manage source code and asset versioning.

Integrations

- **Unity SceneManagement API:** Used for handling level transitions and asynchronous scene loading.
- **Unity UI System:** Integrated for creating responsive menus, HUDs, and interactive dialog panels.

- **Input System:** Handles keyboard and mouse events for character control and UI navigation.
- **(Future Roadmap) Steamworks API:** Planned integration for handling user achievements and cloud save synchronization.